

CALUM HARVEY

07931772321 | calum@gerryharvey.co.uk | [GitHub](#) | [Linkedin](#)

EDUCATION AND QUALIFICATIONS

- Cambridge University** | *PhD Electrical Engineering* | Cambridge, UK **Oct 2022 – Present**
- Joint funded by EPSRC ICASE and Microsoft Research UK
 - Modules taken: Computational Statistics and Machine Learning, and Fibre Optic Communication
 - Built interpretable Gaussian Process-based models to capture nonlinear fibre-optic behaviour while providing calibrated confidence estimates for real-world deployment
 - Designed uncertainty-aware optimisation methods that increase robustness in dynamic systems
- Warwick University** | *MSc Data Science* | Coventry, UK **Oct 2020 – Oct 2021**
- Dissertation: Developed new Artificial Intelligence algorithm for maximising influence spread in non-uniform large-scale social networks
 - Modules include: Data Mining, Algorithmic Game Theory, Natural Language Processing
- Newcastle University** | *BSc (Hons) Computer Science* | Newcastle, UK **Sep 2017 – Jun 2020**
- Dissertation: Artificial Intelligence algorithm implementation and performance comparison in Python, using data science libraries and MySQL
- Stewart's Melville College** | Edinburgh, UK **Aug 2011 – Jun 2017**
- *Advanced Highers*: **B** Computer Science
 - *Highers*: 7 subjects with **AAB** Maths, Physics, Computer Science

WORK EXPERIENCE AND EXTRACURRICULAR ACTIVITIES

- Microsoft Azure Fibre** | Research Intern | Southampton, UK **Jul 2025 – Dec 2025**
- Developed probabilistic ML models for hollow-core fibre links using PyTorch and GPyTorch, improving robustness and interpretability under real-world noise.
 - Collaborated with senior researchers to refine production modelling frameworks for field deployment
 - Worked across the full modelling lifecycle, from exploratory data analysis through model design, training, and evaluation on real-world measured optical link datasets
 - Designed and validated data-driven models that outperformed traditional physics-based baselines on noisy optical measurements
 - Work carried out during the internship is forming the basis of a journal paper currently in preparation
- Cambridge University Squash Club** | Treasurer | Cambridge, UK **Jul 2023 – May 2025**
- Managed a £20k+ annual budget and built automated financial-tracking tools, ensuring accurate, transparent operations for a 150-member university sports club
- Kainos** | Data Engineer | London, UK **Aug 2021 – Sep 2022**
- Carried out projects utilising both Azure and AWS cloud platforms for public sector organisations
 - Developed automated infrastructure deployment framework using Terraform
 - Implemented production machine learning model tracking utilising Databricks with MLflow

PUBLICATIONS

- **C. Harvey**, M. S. Faruk and S. J. Savory, "Data-Driven Erbium-Doped Fiber Amplifier Gain Modeling Using Gaussian Process Regression," in *IEEE PTL*, vol. 36, no. 18, pp. 1097-1100
- **C. Harvey** and S. J. Savory, "Optical Power Spectrum Prediction Using Cascaded Learning with Uncertainty Propagating Noisy Input Gaussian Processes," in *JLT*, vol. 43, no. 16, pp. 7639-7649
- **C. Harvey**, X. Ye, S. J. Savory, "Uncertainty-Aware OSNR Maximization Through Joint Optimization of Channel Power and EDFA Configuration" (under review)

SKILLS, INTERESTS AND AWARDS

- **Machine Learning**: Gaussian Processes, Bayesian optimisation, uncertainty quantification, signal processing, deep learning, large-scale data processing
- **Programming & Tools**: Python, PyTorch, GPyTorch, NumPy, Pandas, SciPy, Optuna, SQL, Spark, Databricks, MLflow, Terraform, Git, Docker, Azure, AWS
- **Certifications**: Microsoft Certified: Azure Data Fundamentals, Databricks Associate Developer for Apache Spark 3.0
- **Competitions**: Exante trading competition (top 10%), QRT portfolio optimisation quantitative trading competition (ongoing)
- **Interests**: Squash (Cambridge University first team), golf, tennis, reading